

The burden of disease in Italy, 1990-2023: a subnational analysis from the Global Burden of Disease Study 2023

Il carico di malattia in Italia, dal 1990 al 2023: un'analisi regionale dal Global Burden of Disease Study 2023

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Abstract

Objectives: to provide a comprehensive and comparable assessment of life expectancy (LE), health-adjusted life expectancy (HALE), Years Lived with Disability (YLDs), Years of Life Lost (YLLs), and Disability-Adjusted Life Years (DALYs) across Italian regions from 1990 to 2023, identifying temporal, geographical, and sex-specific patterns to inform targeted public health strategies.

Design: descriptive epidemiological study using estimates from the Global Burden of Disease Study (GBD) 2023.

Setting and participants: the analysis covered the Italian population, stratified by sex, five macro-regions, and 21 subnational locations, including 19 Regions and 2 Autonomous Provinces.

Main outcome measures: LE, HALE, all-age and age-standardized rates YLDs, YLLs, and DALYs per 100,000 population; all estimates were presented with their 95% uncertainty intervals.

Results: between 1990 and 2023, LE increased substantially across all regions and in both sexes, despite the temporary decline associated with the COVID-19 pandemic. In 2023, the highest LE was recorded in the North-East, with 81.4 years (95%CI 81.1-81.6) for males and 85.8 years (95%CI 85.6-86.1) for females, while the lowest values were found in the Islands, at 79.7 years (95%CI 79.3-80.1) for males and 84.2 years (95%CI 83.9-84.5) for females. However, differences diminished when considering HALE. YLDs increased steadily over time, reflecting not only population ageing but also the expanding burden of chronic and non-fatal conditions. In contrast, YLLs declined markedly in both sexes, driven particularly by substantial reductions in mortality from neoplasms and cardiovascular diseases. However, in 2023, ischemic heart disease remained the leading contributor to all-age DALYs among males, while Alzheimer's disease was the top cause among females. Anxiety disorders represented the leading cause of age-standardized DALYs in females and showed marked increases since 1990. Among males, ischemic heart disease continued to appear as the primary cause of age-standardized DALYs.

Conclusions: health in Italy improved substantially over the past three decades, primarily due to declining premature mortality. However, the burden of disability increased, particularly among women, and important geographical

What is already known

- In recent decades, Italy has undergone profound demographic and epidemiological changes, characterised by an ageing population and changes in disease patterns.
- Despite high life expectancy, geographical inequalities between the North-Centre and South&Islands still persist today.
- Non-communicable diseases are the main source of disease burden and pressure on the health system.

What this study adds

- Between 1990 and 2023, life expectancy increased in all Italian regions and for both sexes, but sex and regional gaps persist, although these narrow and disappear when healthy life expectancy is considered.
- Ischaemic heart disease remains the leading cause of DALYs in males, while Alzheimer's disease and other dementias emerge as the main drivers in females.
- The results highlight a shift in the burden of disease towards chronic conditions characterised by a high disability burden, with significant implications in terms of prevention and long-term care.

and sex-specific inequalities persist. Strengthening prevention policies, addressing behavioural and metabolic risk factors and mental health, improving long-term care, and promoting healthy ageing are essential to reducing health disparities.

Keywords: burden of disease, health inequalities, Global Burden of Disease Study.

Riassunto

Obiettivi: fornire una valutazione comprensiva e comparabile dell'aspettativa di vita alla nascita (LE), dell'aspettativa di vita in salute (HALE), degli anni di vita vissuti con disabilità (YLDs), degli anni di vita persi per morte prematura (YLLs) e degli anni di vita vissuti aggiustati per disabilità (DALYs) nelle regioni italiane dal 1990 al 2023, identificando andamenti temporali, differenze geografiche e di genere al fine di orientare le strategie di salute pubblica.

Disegno: studio epidemiologico descrittivo basato sulle stime del Global Burden of Disease Study (GBD) 2023.

Setting e partecipanti: l'analisi si concentra sulla popolazione italiana, stratificata per sesso, per cinque macroregioni e per 21 unità subnazionali, comprendenti 19 Regioni e 2 Province Autonome.

Principali misure di outcome: LE, HALE, tassi grezzi e standardizzati per età di YLDs, YLLs, and DALYs per 100,000 individui; tutte le stime sono accompagnate dai relativi intervalli di incertezza al 95%.

Risultati: tra il 1990 e il 2023, l'aspettativa di vita è aumentata in modo significativo in tutte le regioni e per entrambi i sessi, nonostante la temporanea diminuzione verificatasi durante la pandemia di COVID-19. Nel 2023, i valori più elevati di aspettativa di vita si registrano nel Nord-Est, con 81,4 anni (IC95% 81,1-81,6) per i maschi e 85,8 anni (IC95% 85,6-86,1) per le femmine, mentre i valori più bassi si rilevano nelle Isole, con 79,7 anni (IC95% 79,3-80,1) nei maschi e 84,2 anni (IC95% 83,9-84,5) nelle femmine. Tuttavia, le differenze geografiche e di sesso si attenuano sensibilmente quando si considera l'aspettativa di vita in salute. Gli YLDs sono aumentati in modo marcato nel tempo in tutte le regioni, riflettendo sia l'invecchiamento della popolazione sia il crescente peso delle condizioni croniche e non fa-

tali. Al contrario, gli YLLs sono diminuiti sostanzialmente in entrambi i sessi, trainati soprattutto da riduzioni significative della mortalità per tumori e malattie cardiovascolari. Nel 2023, la cardiopatia ischemica rimane la principale causa di DALYs a tutte le età nei maschi, mentre la malattia di Alzheimer rappresenta la prima causa nelle femmine. I disturbi d'ansia costituiscono, invece, la principale causa di DALYs standardizzati per età nelle femmine, mostrando aumenti rilevanti dal 1990. Nei maschi, invece, la cardiopatia ischemica continua a essere la prima causa di DALYs standardizzati per età.

Conclusioni: la salute della popolazione italiana è migliorata sensibilmente negli ultimi trent'anni, grazie soprattutto alla riduzione della mortalità prematura. Tuttavia, il carico di disabilità è aumentato, in particolare tra le donne, e persistono importanti disuguaglianze geografiche e di genere. Per ridurre queste disparità, è fondamentale rafforzare le politiche di prevenzione, affrontare i fattori di rischio comportamentali e metabolici, intervenire sui disturbi mentali, potenziare l'assistenza a lungo termine e promuovere un invecchiamento in buona salute.

Parole chiave: carico di malattia, disuguaglianze di salute, Global Burden of Disease Study

Background

Over the past three decades, Italy has undergone profound demographic and epidemiological transformations, characterized by sustained population ageing, declining fertility, and marked shifts in the distribution of diseases and risk factors. This pattern represents a critical challenge from economic, social, and political perspectives. In 2023, Italy was the oldest country in Europe, with a mean age of 48.4 years, one of the lowest birth rates (7 per 1,000 population), and a fertility rate of 1.2 children per woman, well below the replacement level and approaching its historical minimum.¹ Moreover, although life expectancy (LE) at birth in Italy has long been among the highest in Europe, substantial regional inequalities persist, particularly between the northern and central regions and the South and Islands.² At the same time, the burden of non-communicable diseases (NCDs), including cardiovascular diseases, cancers, diabetes, musculoskeletal disorders, and neurodegenerative conditions, has progressively increased, reshaping the health needs of the population and exerting growing pressure on the national health system. In particular, data from the "Progressi delle Aziende Sanitarie per la Salute in Italia" (PASSI) surveillance system showed that, between 2023 and 2024, 18% of individuals aged 18-69 years reported having been diagnosed with at least one chronic disease during their lifetime. Among these, chronic respiratory diseases were the most frequently reported (6%), followed by cardiovascular diseases and diabetes (both 5%).³ The proportion of

people living with at least one chronic condition increased sharply among older adults, reaching 57% in those aged 70 years and over.⁴ In this age group, 27% of respondents reported cardiopathy, 20% reported diabetes, and 13% reported a previous cancer diagnosis. Understanding these dynamics is essential for public health planning, especially in contexts in which population ageing accelerates and where mortality reductions have shifted the burden of disease toward disability rather than early death. However, traditional health indicators often fail to capture this complexity: metrics such as LE provide crucial information on survival, but do not account for the quality of the years lived, as they do not incorporate the substantial impact of non-fatal conditions that increasingly shape population health in high-income settings.

In this framework, the Global Burden of Disease (GBD) Study offers a comprehensive and standardized approach for quantifying health loss across diseases, injuries, risk factors, and populations, allowing the simultaneous assessment of mortality and disability and enabling meaningful comparisons across locations, age groups, sex, and time.

This study presents the most comprehensive subnational analysis to date of life expectancy, health-adjusted life expectancy, years lived with disability, years of life lost, and disability-adjusted life years in Italy from 1990 to 2023, stratified by sex and subnational area, and represents a natural extension of previous work that covered the period up to 2021.² Accordingly, comparisons are presented over the full

1990–2023 period to characterize the overall trajectory of disease burden using the most recent available estimates, as pre-pandemic trends and the effect of COVID-19 on the burden of disease have already been examined in a previous work.²

By jointly examining temporal trends, leading causes, and geographical and sex-specific patterns, we aim to offer an evidence base that can inform equitable and targeted public health strategies.

Methods

Overview

Italy contributes 1,661 data sources to the GBD 2023 Study, including vital registration systems, national surveys, disease registries, and administrative datasets. These sources underpin all GBD estimates and are fully documented in the GBD 2023 Sources Tool, available from: <https://ghdx.healthdata.org/gbd-2023/sources>. The study adheres to GATHER reporting standards, and detailed methodological descriptions are provided in the GBD 2023 capstone papers.^{5–7}

GBD measures

In this analysis, the burden of disease was assessed using the following metrics: life expectancy at birth (LE), health-adjusted life expectancy (HALE), years lived with disability (YLDs), years of life lost (YLLs), and disability-adjusted life years (DALYs).

Specifically, HALE extends the concept of LE by capturing quality of life in addition to length of life. It represents the average number of years an individual can expect to live in full health and is calculated using age-specific mortality rates together with per-capita YLDs. YLDs are estimated by combining disease prevalence with corresponding disability weights that reflect the severity of different health states. On the other hand, YLLs are calculated by multiplying the number of deaths at each age by a reference LE that is identical for males and females. This reference LE is derived from the lowest age-specific mortality rates observed across countries and reflects the normative assumption that individuals of both sexes are entitled to the same standard of potential longevity. Finally, DALYs, defined as the sum of YLLs and YLDs, quantify overall health loss by integrating the contributions of premature mortality and non-fatal health outcomes. YLDs, YLLs, and DALYs were reported as all-age and age-standardized rates per 100,000 population, with 95% uncertainty intervals (UIs). UIs were derived as the 2.5th and 97.5th percentiles of 250 posterior draws and incorporate multiple sources of uncertainty, including variability in input data, measurement error, model specification, and parameter estimation. As such, the resulting UIs represent the combined

uncertainty surrounding each estimate rather than sampling variability alone. Age-standardization was performed using the GBD standard population, derived from the estimated global population by age (World Population Prospects 2012 revision). The GBD standard age structure was obtained by computing the simple arithmetic mean of all country-level age distributions. The GBD classifies health conditions in a hierarchy of four levels, with increasing levels of detail, starting from three level 1 groups comprising:

- communicable, maternal, neonatal, and nutritional diseases;
- non-communicable diseases;
- injuries.

To ensure the highest level of granularity, this work will focus on estimates for level-4 causes.

Data presentation

Estimates were reported for five macro-regions (North-West, North-East, Centre, South, and Islands), and for 21 subnational locations (19 Regions and 2 Autonomous Provinces – APs), following the NUTS-1 and NUTS-2 classification.⁸

Trends in LE, HALE, and all-cause rates of DALYs, YLDs, and YLLs from 1990 to 2023 were analysed separately for males and females. Temporal changes between 1990 and 2023 were assessed using percent changes and their corresponding 95% UIs, as directly estimated within the GBD framework. For cross-sectional comparisons across sex and macro-regions, differences were evaluated by examining the extent of overlap between 95% UIs. When the 95% uncertainty intervals overlapped, this was interpreted as insufficient evidence to support the presence of a substantial difference among groups. This approach was adopted as a descriptive criterion to characterize potential differences, in the absence of formal hypothesis testing.

For each macro-region and sex, the ten level-4 leading causes of YLDs, YLLs, and DALYs in 2023 were identified. For each cause, we reported the corresponding burden and the percent change since 1990, to distinguish true changes in disease burden from ranking dynamics.

Statistical analysis

The analysis followed two main paths: first, the assessment of the actual burden of disease through all-age rates, which allowed for the assessment of the burden of disease without considering the effect of different population structures in different regions; second, the evaluation of health status across years, sexes, and macro-regions using age-standardized rates to allow for comparisons to be drawn across macro-regions as if they had the same age structure, thus adjusting for differences in population age structure.

Results

Life expectancy at birth

Between 1990 and 2023, LE at birth showed an upward trend across all Italian regions, despite the marked shock caused by the COVID-19 pandemic in 2020 (Figure 1). Sex differences were persistent throughout the period, with females consistently showing higher LE than males. A stable geographical gradient also remained, with lower LEs in the South and the Islands (Table S1, online Supplementary Materials).

In 2023, the highest LE was recorded in the North-East, with 81.4 years (95%UI 81.1–81.6) for males and 85.8 years (95%UI 85.6–86.1) for females. Conversely, the lowest values were found in the Islands, at 79.7 years (95%UI 79.3–80.1) for males and 84.2 years (95%UI 83.9–84.5) for females.

However, when considering HALE, the sex gap narrowed substantially (Figure S1), and the differences observed for LE were no longer evident once uncertainty was taken into account. Similarly, the geographical gradient was also attenuated after considering the uncertainty around the estimates (Table S2).

All-age burden of disease

Figure 2 illustrates the decomposition of all-age DALYs in Italy for males and females. Among males, YLLs consistently represented most of the burden, whereas in females, the contribution of YLLs and YLDs remained more evenly distributed throughout the period. Specifically, the proportion of YLLs within total DALYs decreased from 68.2% to 59.0% in males, while in females it increased slightly from 46.3% to 51.4% (Figure 3).

In 2023, all-age DALY rates for all causes in males were 34,326.2 (95%UI 30,843.3–38,121.3) in the Islands, 32,700.5 (95%UI 29,544.9–36,273.0) in the South, 32,489.9 (95%UI 29,207.1–36,368.7) in the Centre, 31,920.3 (95%UI 28,637.5–35,715.1) in the North-East and 32,480.0 (95%UI 29,498.2–36,137.3) in the North-West (Table 1). Among females, DALY rates were 32,825.2 (95%UI 28,698.0–37,628.2) in the Islands, 31,372.7 (95%UI 27,413.3–35,798.7) in the South, 32,381.3 (95%UI 28,364.9–37,312.5) in the Centre, 31,693.0 (95%UI 27,693.0–36,207.3) in the North-East and 32,866.6 (95%UI 29,091.0–37,659.8) in the North-West. Despite geographical differences in point estimates, the overlap of 95% UIs suggested limited evidence for clear inequalities.

All-age DALYs decreased considerably over time among males only in the North-West, North-East and Centre, whereas they remained essentially stable in the South and Islands (Figure 4a). On the other hand, among females, DALY rates in 2023 were substantially higher compared to 1990 in all macro-regions except the North-East, where they remained largely unchanged (Figure 4b).

Analysis of all-age YLDs showed consistently higher rates in females than in males across all macro-regions (Figure S2). In both sexes, YLDs increased steadily over time, especially in the South and Islands; however, by 2023, YLD rates were comparable across macro-regions for both males and females (Table S3). Conversely, YLLs decreased substantially among males in all macro-regions since 1990, whereas in females, reductions were limited to the North-West, North-East, and Centre. By contrast, female YLLs increased over time in the South and Islands (Figure S3, Table S4).

Regarding the leading causes of DALYs in 2023, ischaemic heart disease was the primary contributor among males in all macro-regions, with rates of 2,439.6 (95%UI 2,198.4–2,662.7) in the North-West, 2,326.0 (95%UI 2,071.1–2,566.2) in the North-East, 2,604.9 (95%UI 2,340.9–2,875.8) in the Centre, 2,890.4 (95%UI 2,652.6–3,142.3) in the South, and 2,717.9 (95%UI 2,491.3–2,967.9) in the Islands (Figure S4, Table S5). Among females, Alzheimer's disease and other dementias was the leading cause of DALYs throughout the country, with rates of 1,109.3 (95%UI 505.2–2,306.1) in the North-West, 1,160.3 (95%UI 536.5–2,408.6) in the North-East, 1,127.2 (95%UI 511.8–2,347.3) in the Centre, 1,321.6 (95%UI 600.2–2,852.3) in the South, and 1,330.4 (95%UI 614.1–2,881.8) in the Islands (Figure S5, Table S6).

The leading causes of all-age YLDs are reported in Figure S6 and Table S7 for males and in Figure S7 and Table S8 for females. For YLLs, results are shown in Figure S8 and Table S9 for males and Figure S9 and Table S10 for females.

Age-standardized burden of disease

As shown in Figure S10, the decomposition of age-standardized DALYs in Italy has changed substantially over time, with distinct patterns for males and females. Among males, YLLs consistently represented the predominant share of total DALYs; however, their contribution declined markedly throughout the period, decreasing from 67.5% in 1990 to 51.2% in 2023, resulting in a more even distribution between YLLs and YLDs (Figure S11). In females, YLLs were already the smaller component in 1990, accounting for 49.2% of total DALYs, and their share further declined to 34.1% by 2023. In both sexes, reductions in mortality were not accompanied by a comparable compression of morbidity, resulting in a growing relative contribution of YLDs to the overall burden of disease.

Age-standardized DALY rates declined substantially over time for both sexes in all macro-regions, reflecting long-term improvements in population health (Figure S12). A marked spike is observed in 2020–2021, corresponding to the COVID-19 pandemic, after which rates partially returned toward the pre-pandemic trajectory. In 2023, the South and Islands con-

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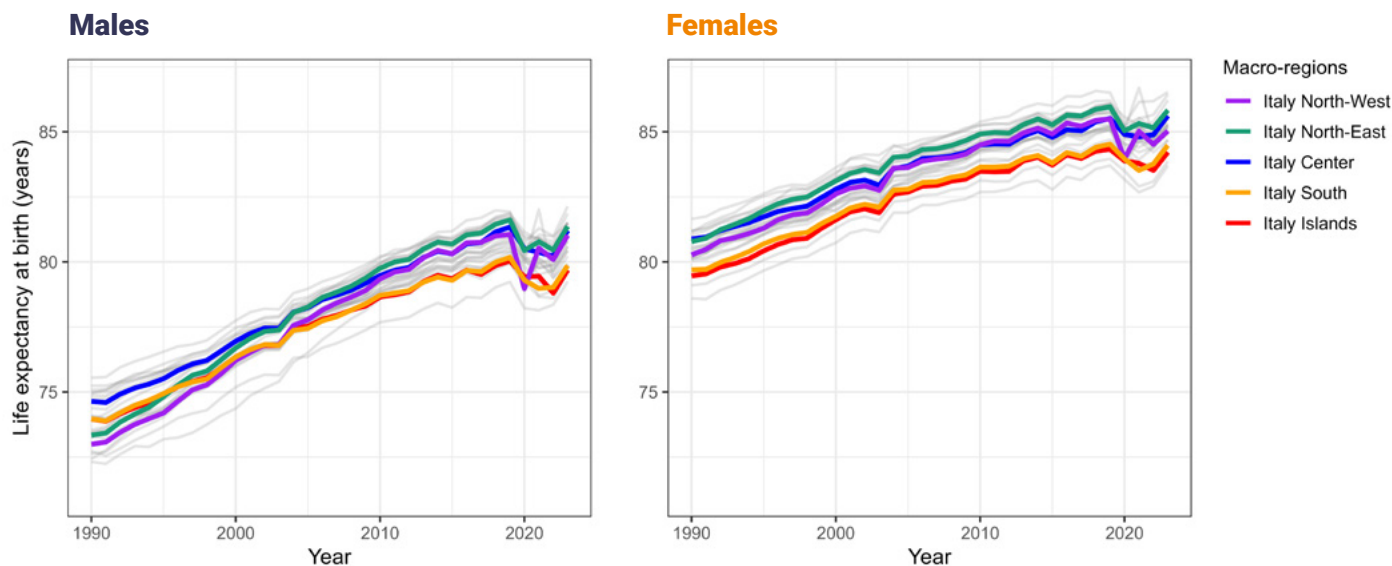


Figura 1. Andamento dell'aspettativa di vita alla nascita dal 1990 al 2023 nei maschi e nelle femmine, per macroregione. Le linee grigie rappresentano gli andamenti delle singole regioni.

Figure 1. Trend in life expectancy at birth from 1990 to 2023 in males and females, by Italy macroregion. Grey lines represent the trends of individual regions.

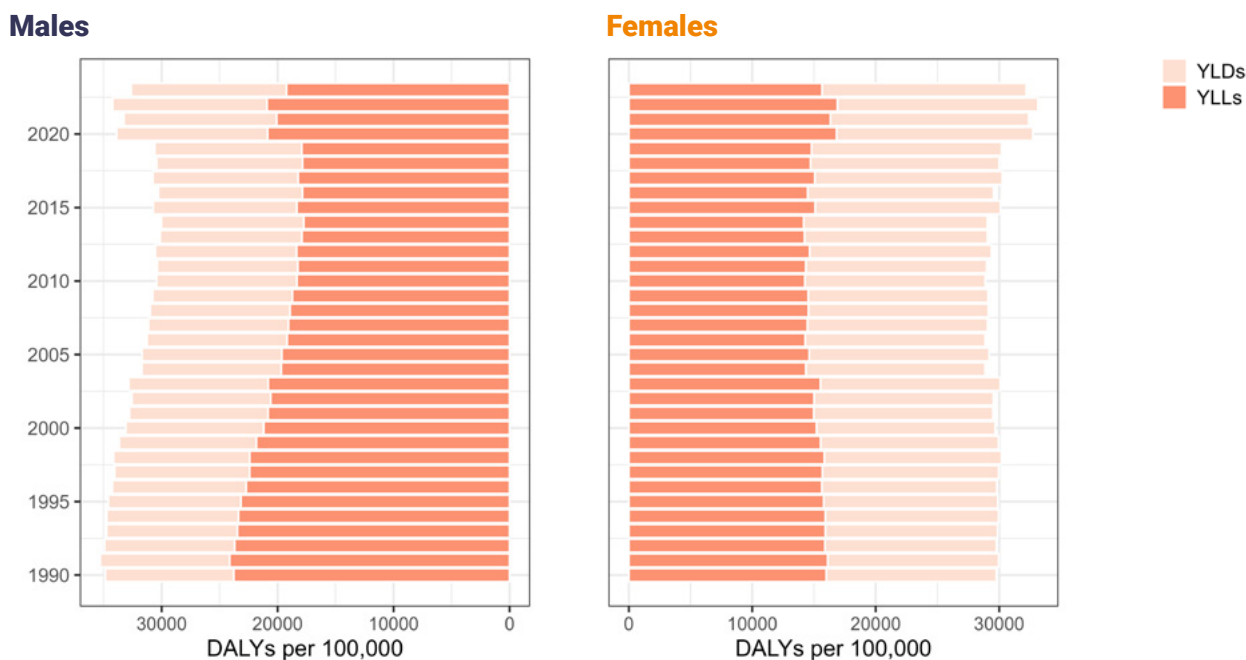


Figura 2. Anni di vita vissuti aggiustati per disabilità (DALYs) per tutte le età nel 2023 in Italia, per maschi e femmine, scomposti in anni di vita persi per morte prematura (YLLs) e anni di vita vissuti con disabilità (YLDs).

Figure 2. All-age Disability-Adjusted Life Years (DALYs) in Italy in 2023, for males and females, and decomposed into years of life lost due to premature mortality (YLLs) and years lived with disability (YLDs).

sistently showed higher point estimates suggesting the presence of a geographical gradient; however, overlapping 95% UI indicated limited evidence for clear difference (Table S11).

Regarding YLDs, rates remained relatively stable across all regions and for both sexes up to 2019, followed by a marked rise in 2020 and the subsequent years (Figure

S13). By 2023, YLD levels were not different from those observed in 1990 for males across all macro-regions, whereas females showed an increase in the North-West and North-East (Table S12). Conversely, YLL rates declined substantially in all regions between 1990 and 2023 for both sexes (Figure S14). The reduction was steeper among males than females, likely reflecting a

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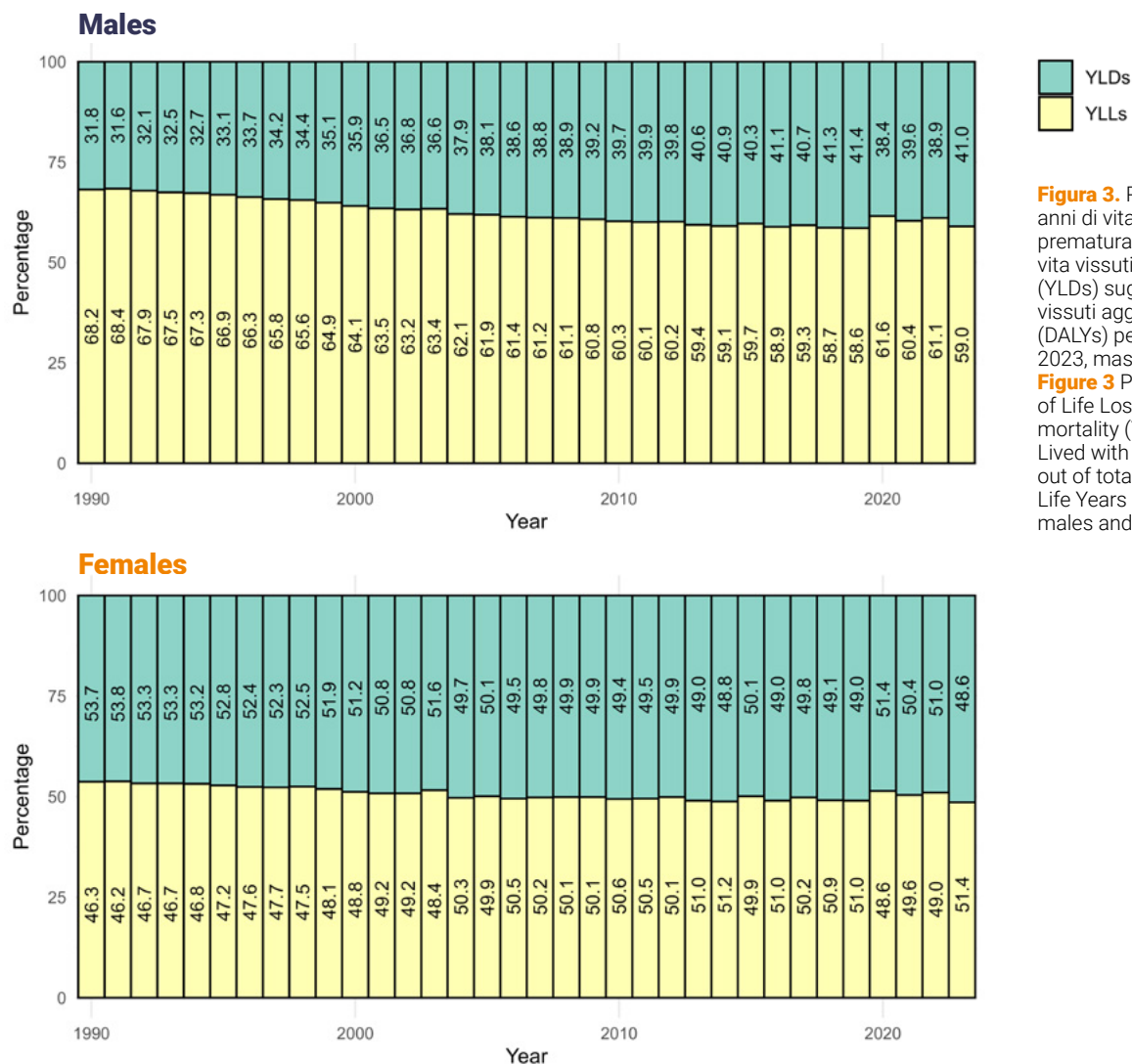


Figura 3. Percentuale di anni di vita persi per morte prematura (YLLs) e anni di vita vissuti con disabilità (YLDs) sugli anni di vita vissuti aggiustati per disabilità (DALYs) per tutte le età. Italia, 2023, maschi e femmine.

Figure 3 Percentage of Years of Life Lost due to premature mortality (YLLs) and Years Lived with Disability (YLDs) out of total Disability-Adjusted Life Years (DALYs). Italy, 2023, males and females.

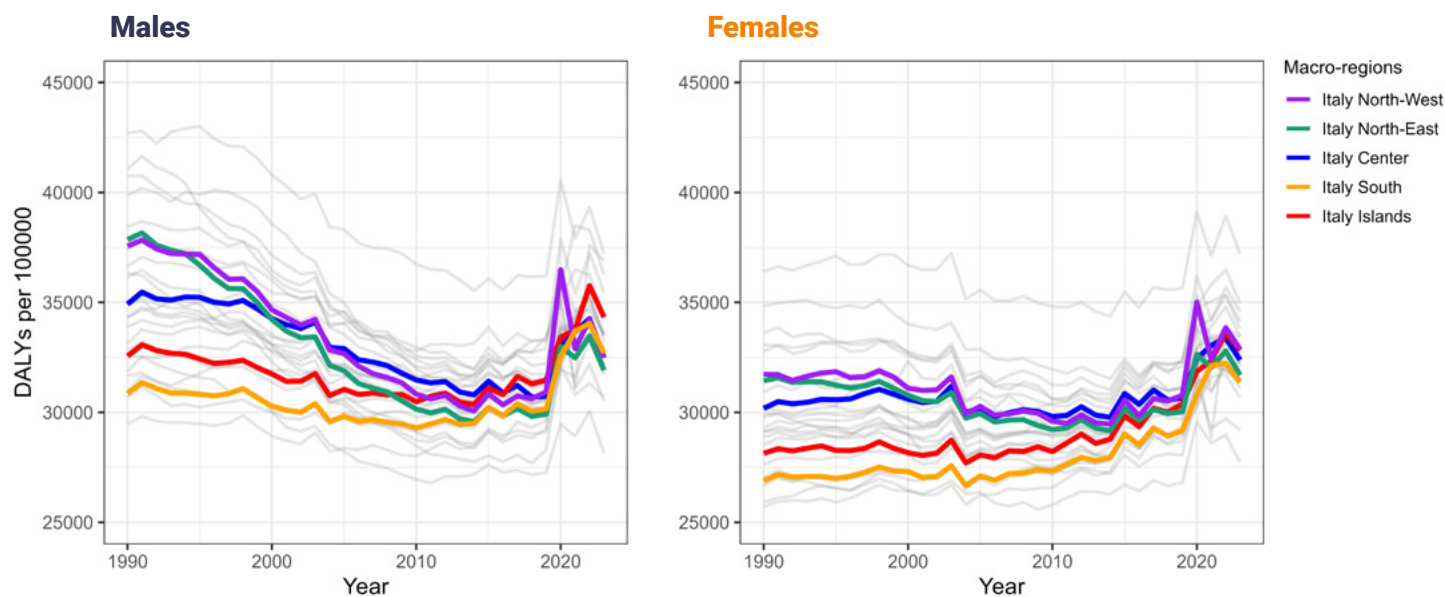


Figura 4. Andamento degli anni di vita vissuti aggiustati per diabilità (DALYs) dal 1990 al 2023 nei maschi e nelle femmine, per macroregione. Le linee grigie rappresentano gli andamenti delle singole regioni.

Figure 4 Trend in all-age Disability-Adjusted Life Years (DALYs) from 1990 to 2023 in males and females, by Italian macroregion. Grey lines represent the trends of individual regions.

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Area	Disability-Adjusted Life Years per 100,000 (95%UI)					
	Males			Females		
	1990	2023	Change (%)	1990	2023	Change (%)
North-Western Italy	37,559.3 (34,730.5; 40,757.9)	32,480.0 (29,498.2; 36,137.3)	-13.5 (-16.7; -10.2)	31,732.3 (28,245.2; 35,775.0)	32,866.6 (29,091.0; 37,659.8)	3.6 (0.9; 6.3)
Piemonte	38,438.3 (35,349.5; 41,795.5)	35,459.9 (32,144.0; 39,313.6)	-7.7 (-12.6; -3.0)	33,076.3 (29,682.3; 37,137.6)	34,573.4 (30,487.7; 39,114.7)	4.5 (0.5; 9.2)
Valle d'Aosta	39,890.1 (36,186.1; 43,738.2)	33,580.6 (29,692.5; 37,958.7)	-15.8 (-22.7; -8.9)	31,752.8 (27,871.2; 36,328.6)	32,249.5 (27,312.8; 37,515.8)	1.6 (-4.7; 8.1)
Lombardia	36,136.1 (33,344.2; 39,153.6)	30,524.5 (27,513.2; 34,169.7)	-15.5 (-19.5; -12.0)	30,167.0 (26,693.1; 34,262.5)	31,488.3 (27,771.8; 36,174.5)	4.4 (1.3; 7.6)
Liguria	42,701.0 (39,106.5; 46,729.7)	37,206.5 (33,488.6; 42,158.7)	-12.9 (-18.1; -7.7)	36,409.8 (32,547.6; 41,407.6)	37,189.7 (32,774.9; 42,580.5)	2.1 (-3.4; 7.6)
North-Eastern Italy	37,852.8 (35,077.3; 41,223.5)	31,920.3 (28,637.5; 35,715.1)	-15.7 (-19.1; -12.2)	31,438.3 (27,904.1; 35,477.4)	31,693.0 (27,693.0; 36,207.3)	0.8 (-2.0; 3.5)
AP Trento	36,339.6 (33,277.9; 39,932.8)	30,693.4 (27,164.9; 34,833.5)	-15.5 (-21.6; -9.6)	29,931.7 (26,397.0; 33,994.4)	29,207.4 (24,864.1; 33,939.2)	-2.4 (-8.4; 3.2)
AP Bolzano	32,599.5 (29,435.1; 36,415.2)	28,131.3 (24,434.4; 32,147.4)	-13.7 (-19.9; -7.2)	27,049.0 (23,842.7; 31,028.6)	27,720.2 (23,710.4; 32,634.2)	2.5 (-3.9; 8.7)
Veneto	34,906.0 (32,008.6; 38,360.4)	31,716.0 (28,374.7; 35,514.4)	-9.1 (-13.2; -4.9)	29,125.6 (25,707.6; 32,988.7)	31,100.0 (27,067.8; 35,490.8)	6.8 (2.7; 10.5)
Friuli Venezia Giulia	40,748.5 (37,532.1; 44,447.6)	34,328.0 (30,623.3; 38,565.4)	-15.8 (-20.7; -10.2)	34,834.2 (30,885.5; 39,208.0)	34,132.6 (29,847.5; 39,260.2)	-2.0 (-7.1; 3.2)
Emilia-Romagna	41,050.1 (37,793.9; 44,836.9)	32,115.8 (28,677.6; 36,175.8)	-21.8 (-25.8; -17.2)	33,613.4 (29,893.5; 38,122.8)	32,458.3 (28,234.6; 37,366.4)	-3.4 (-7.6; 0.5)
Central Italy	34,919.8 (31,727.2; 38,340.9)	32,489.9 (29,207.1; 36,368.7)	-7.0 (-9.9; -4.3)	30,183.6 (26,670.2; 34,114.3)	32,381.2 (28,364.9; 37,312.5)	7.3 (4.0; 10.5)
Toscana	37,391.3 (33,966.7; 41,184.4)	33,240.1 (29,870.1; 37,175.3)	-11.1 (-15.0; -6.9)	32,770.9 (28,853.4; 37,164.2)	33,808.3 (29,435.6; 38,844.6)	3.2 (-1.1; 7.0)
Umbria	36,647.6 (32,758.5; 41,023.8)	33,576.1 (29,923.8; 37,521.1)	-8.4 (-14.7; -2.0)	30,930.5 (27,554.6; 35,253.8)	33,439.4 (29,005.4; 38,273.3)	8.1 (2.0; 14.2)
Marche	34,314.7 (31,238.0; 37,971.8)	32,155.5 (28,540.8; 35,915.8)	-6.3 (-12.4; -0.5)	29,279.2 (25,896.9; 33,166.8)	32,072.9 (27,980.0; 36,824.7)	9.5 (4.6; 14.9)
Lazio	33,118.5 (29,907.5; 36,464.1)	31,939.2 (28,622.4; 35,789.1)	-3.6 (-7.2; 0.7)	28,523.5 (25,072.4; 32,590.1)	31,400.4 (27,529.3; 36,193.7)	10.1 (6.2; 14.2)
Southern Italy	30,859.0 (28,288.8; 33,797.0)	32,700.5 (29,544.9; 36,273.0)	6.0 (2.9; 8.6)	26,883.2 (23,870.7; 30,713.6)	31,372.7 (27,413.3; 35,798.7)	16.7 (13.4; 20.3)
Abruzzo	33,888.2 (30,658.7; 37,377.4)	34,317.4 (30,743.7; 38,333.5)	1.3 (-4.9; 6.7)	29,106.1 (25,740.6; 33,187.0)	32,399.2 (27,851.3; 37,363.6)	11.3 (6.0; 17.0)
Molise	34,327.3 (31,204.9; 37,861.3)	36,254.1 (31,978.8; 40,617.2)	5.6 (-2.2; 13.6)	30,171.2 (26,406.9; 34,170.8)	34,959.4 (30,664.2; 39,470.8)	15.9 (8.2; 24.1)
Campania	30,952.1 (28,250.4; 33,948.4)	31,809.3 (28,669.9; 35,479.4)	2.8 (-1.9; 7.0)	27,032.9 (24,033.1; 30,709.5)	30,657.6 (26,795.5; 35,180.3)	3.4 (9.3; 18.2)
Puglia	29,471.4 (26,928.9; 32,453.0)	32,579.3 (29,137.5; 36,358.7)	+10.5 (5.6; 15.5)	25,709.7 (22,750.4; 29,437.9)	31,572.6 (27,611.4; 35,969.0)	+22.8 (17.6; 27.8)
Basilicata	31,664.7 (28,655.2; 35,110.9)	34,242.8 (30,305.2; 38,476.2)	+8.1 (1.0; 15.5)	27,635.7 (24,359.6; 31,675.4)	32,695.2 (28,529.1; 37,316.3)	+18.3 (11.3; 25.7)
Calabria	30,684.1 (28,109.8; 34,047.1)	33,524.3 (30,267.9; 37,376.4)	+9.3 (3.2; 15.3)	26,669.6 (23,728.1; 30,737.1)	31,486.7 (27,415.7; 35,979.8)	+18.1 (12.5; 23.6)
Italian Islands	32,562.0 (29,789.0; 35,925.9)	34,326.2 (30,843.3; 38,121.3)	+5.4 (1.3; 9.1)	28,138.4 (24,893.2; 32,135.7)	32,825.2 (28,698.0; 37,628.2)	+16.7 (13.2; 20.0)
Sicilia	32,801.6 (29,911.9; 36,198.6)	33,460.8 (30,054.7; 37,378.3)	+2.0 (-3.0; 6.6)	28,866.8 (25,691.6; 32,749.0)	32,632.2 (28,784.9; 37,581.1)	+13.0 (8.9; 17.0)
Sardegna	31,849.8 (28,878.4; 34,906.9)	36,893.7 (32,800.2; 41,084.6)	+15.8 (9.7; 21.9)	25,911.9 (22,691.3; 30,086.1)	33,412.3 (28,956.7; 38,133.7)	+28.9 (23.0; 34.4)

Table 1. All-age Disability-Adjusted Life Year (DALY) rates for males and females across Italian regions and macro-regions. Rates are reported with 95% uncertainty intervals (95%UI).

Tabella 1. Tassi di anni di vita vissuti aggiustati per disabilità (DALYs) per tutte le età per maschi e femmine nelle regioni e macroregioni italiane. I tassi sono riportati con i relativi intervalli di incertezza al 95% (95%UI).

floor effect in females, whose YLL rates were already lower in 1990, potentially due to sex-specific differences in baseline risk factor profiles and their trajectories over time. A clear geographical gradient persisted throughout the period, with higher YLL rates in the South and the Islands (Table S13).

The leading causes of age-standardized DALYs for males are reported in Table S14 and illustrated in Figure S15. Ischaemic heart disease remained the primary contributor to DALYs in all macro-regions, accounting for 1,105.2 DALYs (95%UI 1,007.2-1,203.1) in the North-West, 1,035.3 (95%UI 936.5-1,141.7) in the North-East, 1,126.5 (95%UI 1,022.6-1,226.4) in the Centre, 1,360.4 (95%UI 1,241.9-1,485.0) in the South, and 1,237.6 (95%UI 1,124.2-1,356.3) in the Islands. For females, anxiety disorders were the leading cause of DALYs across all macro-regions (Figure S16), with estimates of 1,365.3 (95%UI 855.2-2,021.8) DALYs in the North-West, 1,346.3 (95%UI 838.3-2,005.3) in the North-East, 1,337.4 (95%UI 835.8-2,005.3) in the Centre, 1,339.5 (95%UI 833.8-1,990.3) in the South, and 1,293.6 (95%UI 806.3-1,941.5) in the Islands. The burden of anxiety disorders has nearly doubled since 1990 in every macro-region (Table S15).

Results for YLDs are presented in Table S16 and Figure S17 for males, and in Table S17 and Figure S18 for females. Corresponding results for YLLs are reported in Table S18 and Figure S19 for males, and in Table S19 and Figure S20 for females.

Discussion

The analysis here presented shows that LE at birth increased substantially in Italy between 1990 and 2023 across all regions and in both sexes, despite the temporary decline associated with the COVID-19 pandemic. However, a persistent sex gap remains, with females consistently showing higher values than males throughout the period. A stable geographical gradient also persists, with the South and Islands exhibiting lower LE than the Centre and the North for both sexes. Notably, when HALE is considered, both the magnitude of the sex difference and the geographical gradient attenuate and ultimately disappear, reflecting the well-described male-female health-survival paradox: women live longer, but spend more years in poor health.⁹ Therefore, the overall increase in LE was mainly due to the reductions in YLL rates, especially those from neoplasms and cardiovascular diseases. For neoplasms, declines in YLLs may reflect major advancements in early detection and treatment. Specifically, screening programmes have increased the likelihood of early identification and management of several cancers, particularly breast, cervical, and colorectal cancers, contributing to improved survival outcomes.¹⁰⁻¹²

Moreover, additional improvements may stem from public health interventions targeting key risk factors, such as smoking cessation campaigns, reductions in environmental and occupational carcinogens, and the implementation of HPV vaccination campaigns.

In general, Italy performs comparatively well in several domains of NCD control, particularly tobacco regulation: the WHO Noncommunicable Diseases Progress Monitor 2025 reports that Italy has implemented all MPOWER-recommended policies for tobacco control at the highest level.¹³ However, the report highlights gaps in other domains, such as limited implementation of policies targeting unhealthy diets and insufficient national action plans to comprehensively address NCD prevention. These mixed performances are consistent with patterns observed in our analysis, where significant progress in mortality reduction coexists with persistent regional inequalities and an important burden of disability. For this reason, strengthening prevention-oriented strategies, especially those addressing metabolic and behavioural risk factors, remains essential.

In 2023, all-age DALY rates were comparable across macro-regions and sexes, although the composition of the burden differed markedly by sex. In males, YLLs dominated the burden throughout the study period, although their contribution declined over time, reflecting progresses in reducing premature mortality.^{14,15} In females, DALYs were more evenly distributed between YLLs and YLDs. Overall, trends in all-age YLDs indicate a substantial increase over time across most macro-regions, consistent with population ageing and the rising prevalence of chronic, non-fatal conditions. In contrast, all-age YLLs declined markedly in both sexes and across all regions, reflecting sustained reductions in premature mortality. These opposing patterns highlight the epidemiological transition in Italy, where gains in survival have shifted the burden of disease toward disability rather than early death. Moreover, the analysis of leading causes of DALYs highlights marked sex-specific differences. Among males, ischaemic heart disease remained the primary contributor across all macro-regions, despite a substantial reduction in its burden since 1990. In contrast, among females, Alzheimer's disease and other dementias emerged as the leading cause of DALYs in 2023, with burdens that more than doubled over the study period.

The rapid increase of Alzheimer's disease, observed in both sexes albeit with different magnitudes, is largely attributable to population ageing, increased survival at older ages, and the limited availability of disease-modifying treatments. Improved awareness and diagnostic capacity may have also contributed to higher recorded prevalence, particularly in regions with stronger primary care and specialist services. Evidence suggests

that the risk of cognitive decline and dementia can be mitigated through modifiable behavioural and metabolic factors, including regular physical activity, smoking cessation, moderation in alcohol consumption, and weight control.¹⁶ Therefore, our findings reinforce the need to strengthen long-term care, expand community-based services, and enhance caregiver support nationwide, while underscoring the importance of comprehensive prevention strategies alongside improved care for affected individuals.

It is important to consider that, within the GBD framework, uncertainty intervals reflect multiple sources of model-based uncertainty. Consequently, the absence of statistically significant differences among the considered macro-regions should not be interpreted as evidence of homogeneous health needs, but rather as an indication that such differences warrant further, area-specific investigation. Persistent differences in point estimates, consistent geographical gradients, and stable ranking patterns over time may still indicate meaningful disparities with important implications for health system planning and resource allocation.

A further notable finding concerns type-2 diabetes mellitus. Among males, diabetes ranked within the top ten causes of all-age DALYs in all macro-regions, rising from ninth position in the North-West to second in the Islands. This gradient is largely driven by YLDs, which were almost three times higher in the Islands than in the North-West. In females, diabetes entered the top ten only in the South and Islands. This pattern mirrors the geographical distribution of risk factors across Italy, especially obesity and sedentary behaviour, which show a higher prevalence in the South and Islands, as highlighted in the Osservasalute report 2023.¹⁷ For this reason, targeted prevention strategies addressing such disparities are crucial. As for all-age YLDs, low back pain emerged as the leading cause in both sexes, in line with other GBD findings showing this condition as a primary driver of disability worldwide.¹⁸ This finding underscores the importance of comprehensive strategies to promote physical activity, weight management, injury prevention, and adequate access to rehabilitation services. For YLLs, ischaemic heart disease remained the dominant contributor in males across all macro-regions, whereas among females it was the leading cause only in the South, with Alzheimer's disease ranking first in the other macro-regions.

Age-standardized DALY rates decreased markedly in both sexes over time, confirming improvements in population health independent of demographic ageing. However, while age-standardized YLDs remained stable in males across all regions, they increased markedly among females in the North-West and North-East, suggesting a rising burden of disa-

bility among women in these regions that cannot be explained by ageing alone. Conversely, age-standardized YLLs declined sharply in all macro-regions and in both sexes, indicating sustained reductions in premature mortality. Finally, while ischaemic heart disease remained the leading cause of age-standardized DALYs in males, anxiety disorders emerged as the leading contributor among females, with burdens nearly doubling since 1990. This trend aligns with recent reports documenting rising mental health needs globally, especially among women and common mental health disorders, and underscores the critical importance of expanding mental-health support, strengthening early identification, and reducing barriers to treatment, in particular enhancing access at the primary care level.¹⁹ The lower burden observed in males may also indicate underdiagnosis, given the underestimation of anxiety symptoms and gender-related stigma surrounding mental health.^{20,21} For this reason, gender-sensitive prevention and facilitated care pathways, alongside real-time mental health surveillance, are essential to curb this tendency.

This analysis has limitations, some inherent to estimation processes and others described in the GBD 2023 capstone papers.⁵⁻⁷ First, disability weights used to calculate YLDs may not fully capture cultural and contextual variations in perceived health loss. Moreover, the accuracy of results depends on data quality: although Italy has substantial and relatively high-quality data, important gaps remain. Conditions managed primarily in outpatient settings or through local mental-health services may be underrepresented. Moreover, outpatient hospital data are not ICD-coded and are therefore more difficult to incorporate into GBD estimates. In addition, general practitioners and paediatricians use systems not integrated with hospital or mortality databases. Last, national surveillance data are not fully accessible to the GBD and official mortality data are released with a two-year lag, which affects timeliness. Therefore, all these gaps contribute to wider uncertainty intervals for some conditions.

Conclusions

This study moves beyond mortality-centric assessments by documenting how regional convergence in life expectancy masks widening inequalities in disability burden. The comprehensive assessment of LE, HALE, YLDs, and YLLs across Italy underscores the need for a multifaceted public health strategy that addresses persistent sex- and geographically-based inequalities and supports the organization of integrated care pathways spanning health promotion, prevention, diagnosis, treatment, rehabilitation, and palliative care, in order to better respond to popu-

lation needs and improve overall population health. Strengthening preventive interventions, ensuring equitable access to high-quality care, promoting healthy ageing, and tackling the social and economic determinants of health will be essential to reducing the burden of disease nationwide.

In this epidemiological context, the GBD framework represents an added value through its capacity to model and quantify the burden of disability, which is a dimension of population health that remains largely underrepresented in conventional indicators and routinely collected health statistics. However, in high-income settings such as Italy, which have undergone advanced epidemiological and demographic transitions,

disability – rather than mortality – represents the major unresolved challenge. By providing comparable, comprehensive estimates of non-fatal health loss, the GBD enables a deeper understanding of the conditions driving reduced quality of life and functional decline, thereby offering critical insights to inform public health priorities in societies where ageing and chronic disease have become dominant.

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